Job Title: AI Engineer

Location: [Remote / Hybrid / Onsite - City, State]

Job Type: Full-time

Experience Level: Mid-to-Senior

Job Summary

We are seeking a talented and driven **AI Engineer** to design, build, and deploy production-grade artificial intelligence systems. In this role, you will bridge the gap between foundational AI models and real-world software infrastructure to deliver tangible business metrics. You will own the development of end-to-end applications, specifically focusing on Large Language Models (LLMs), semantic search, and autonomous multi-agent frameworks.

Key Responsibilities

- **Architect AI Workflows**: Design and implement application logic using orchestration frameworks like LangChain, LlamaIndex, or LangGraph.
- **Optimize RAG Pipelines**: Build advanced retrieval systems utilizing vector databases like Pinecone, Weaviate, or pgvector.
- **Manage Model Context**: Implement Model Context Protocols (MCP) to securely connect AI models with external tools, databases, and APIs.
- **Evaluate & Fine-Tune**: Develop testing harnesses to rigorously measure accuracy, latency, hallucinations, and instruction-following quality.
- **Deploy to Production**: Maintain MLOps pipelines and orchestrate cloud inference infrastructure on AWS, Google Vertex AI, or Azure ML.
- **Collaborate Cross-Functionally**: Work with software engineers, product managers, and data scientists to scope features and scale data ingestion pipelines.

Required Skills & Qualifications

- **Programming Languages**: Expert-level proficiency in Python, including object-oriented design and unit testing frameworks.
- **AI & Framework Ecosystems**: Hands-on experience with PyTorch or TensorFlow, alongside Hugging Face toolsets.

- **Prompt & Context Engineering**: Mastery of systematic prompt design, few-shot conditioning, and data chunking strategies.
- **Data & Cloud Systems**: Strong command of SQL, RESTful APIs, Docker containerization, and cloud infrastructure management.
- **Education & Experience**: Bachelor's degree in Computer Science, Data Science, or a related field (or equivalent portfolio of publicly deployed AI users).

Nice-to-Haves

- Experience with custom foundation model fine-tuning via techniques like LoRA, SFT, or Direct Preference Optimization (DPO).
- A track record of open-source contributions to popular machine learning repositories.
- Deep understanding of data privacy, safety boundaries, and alignment methods for deployed AI systems.

What Success Looks Like (First 90 Days)

- **Day 30**: Fully audit our existing AI pipelines, set up your local development environment, and ship your first optimization patch to production.
- **Day 60**: Own a core feature branch, design a multi-step reasoning workflow or tool-calling agent, and stand up an automated evaluation suite.
- **Day 90**: Successfully deploy a scalable RAG or agentic system that measurably decreases system latency and reduces downstream hallucinations.